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ISDA 141 - Information Visualization

Evolution of Global Health Risks and the Influence of Environmental Factors

(1990–2023)

DATASET:

This project explores **how environmental risks**, such as air and water pollution, **contribute to the global burden of disease**. By analyzing Disability-Adjusted Life Years (DALYs) across countries and regions, the project aims to understand how environmental factors influence global health patterns over time.

IHME **Institute for Health Metrics and Evaluation**

GLOBAL Burden of Disease (GBD) 2023 Study **Table 13**

The dataset can be found at:

https://ghdx.healthdata.org/record/ihme-data/gbd-2023-yld-daly-hale-risk-1990-2023

Problem Statement

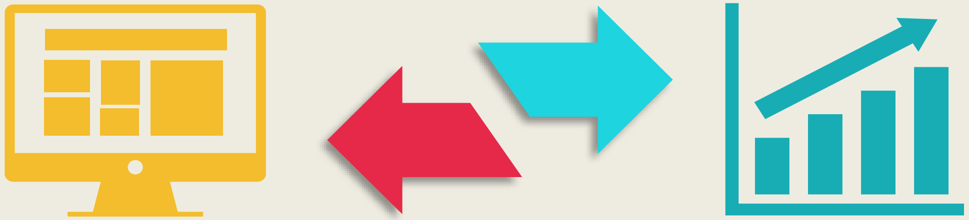
Environmental risks are known contributors to illness and premature death, but their specific impact on population health across regions remains unclear. The central research question guiding this project is:

Research Question

How do environmental risks contribute to overall health problems across different countries and regions?

This question matters because identifying major contributors to global disease burden can guide public health efforts, resource allocation, and prevention strategies. The visualizations created for this analysis help reveal trends, patterns, and regional disparities related to environmental risks.

Dataset Summary



Supplementary Table 13 from the Global Burden of Disease (GBD) 2023 Study

The dataset used for this project comes from a regular release by the Institute for Health Metrics and Evaluation (IHME). Supplementary Table 13 reports Disability-Adjusted Life Years (DALYs) across global locations from 1990 to 2023. The original file contains **595,024 records and 35 attributes**, and after filtering for relevant environmental risk categories and cleaning the data, the working dataset includes **39,668 records and 13 attributes**.
Link: https://ghdx.healthdata.org/record/ihme-data/gbd-2023-yld-daly-hale-risk-1990-2023

This dataset is appropriate because DALYs measure the combined impact of illness and premature death, allowing the visualizations to show how environmental risks affect population health worldwide.

Environmental risks cause millions of lost healthy years worldwide.
This dataset shows how the global disease burden impacts population health.

Visualization

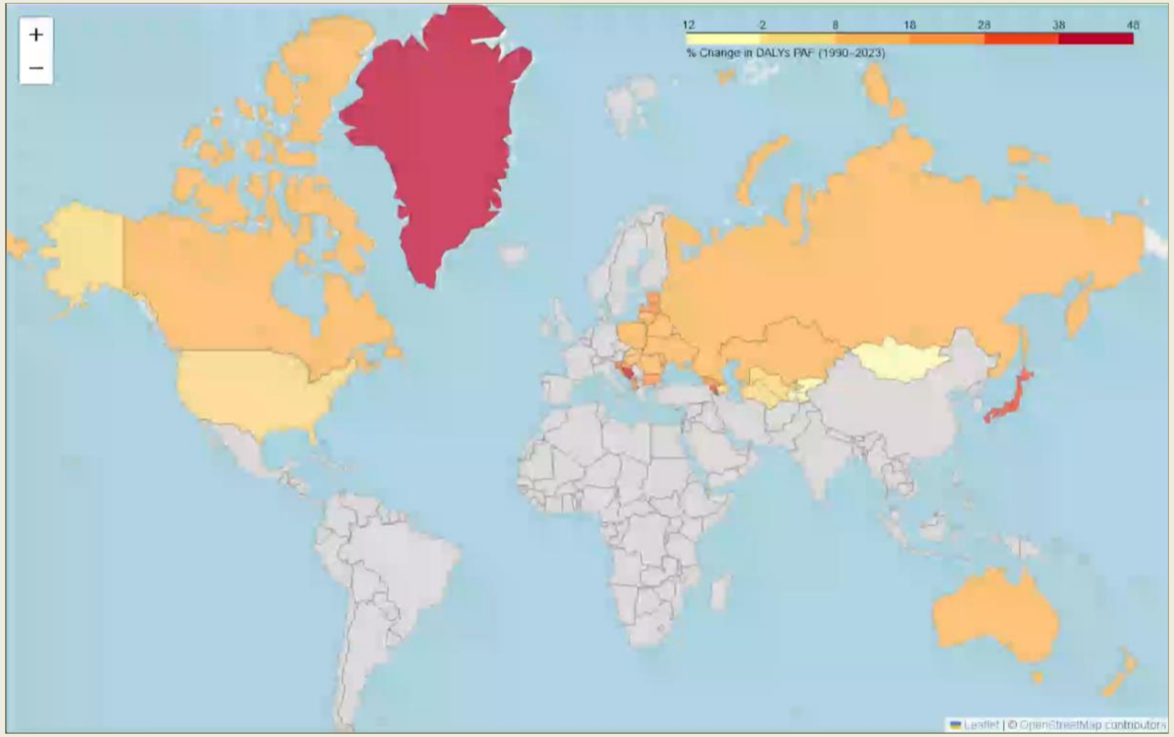
Global Choropleth Map

The color-coded map represents the percentage change in environmental risk contributions to the total disease burden for each country. Performance across different locations is easily identifiable.

Interactivity

Tooltips:

- 1990 PAF value
- 2023 PAF value
- % change



Hover the mouse over the video to show controls

Key findings

Overall, disease-burden contributions related to environmental risks have decreased across the globe, with the exception of a few localized areas.

Visualization

Stacked Area Chart

The height of each stacked layer shows which groups contribute most to the environmental disease burden, providing a clear view of overall trends.

Interactivity

Drop-down menu:

- Displays a red reference line that isolates the selected risk, making it easy to compare and focus on one category at a time.

Hover the mouse over the video to show controls

Key findings

Over the last 30 years, health disease burdens have decreased, including those linked to environmental risks. However, their share of the total burden has remained fairly constant.

Visualization

Scatter Plot

• Compares the environmental burden to the total health burden across regions and countries, from 1990 to 2023.

Interactivity

Animation:

- Shows how values change over the 30-year period, helping viewers better understand long-term trends.

Tooltips:

- Display the actual values for both Environmental and Total burden, giving viewers a clearer and more detailed understanding.

Key findings

Highlights that the environmental burden consistently represents only a small portion of the total health burden in every country, while both have decreased proportionally over time.

Hover the mouse over the video to show controls

Visualization

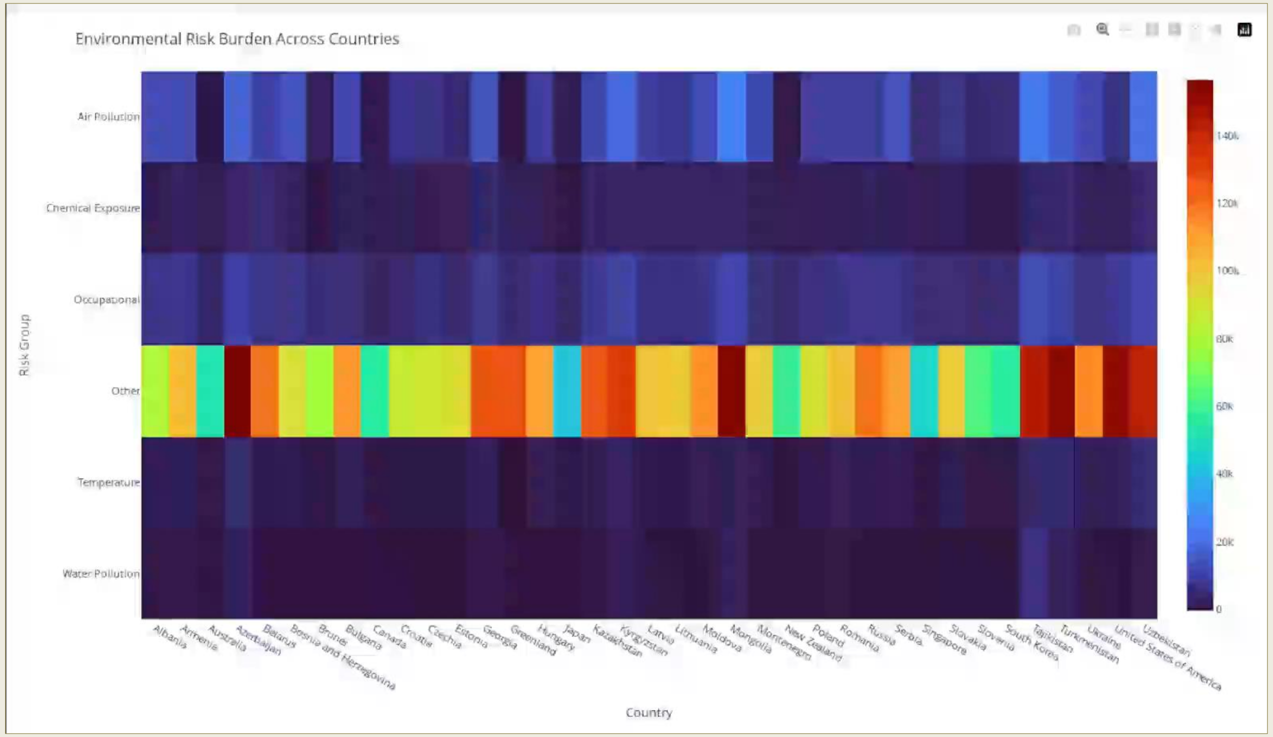
Heatmap

Allows users to quickly see which countries have the highest environmental burden, based on different risk groups and strong color intensities.

Interactivity

Tooltips:

- Shows each location and its risk groups, helping users identify which groups are most prevalent.



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Key findings

Some smaller countries still have high environmental burdens, showing that pollution arises from different factors and that solutions need to be tailored to each place.

Visualization

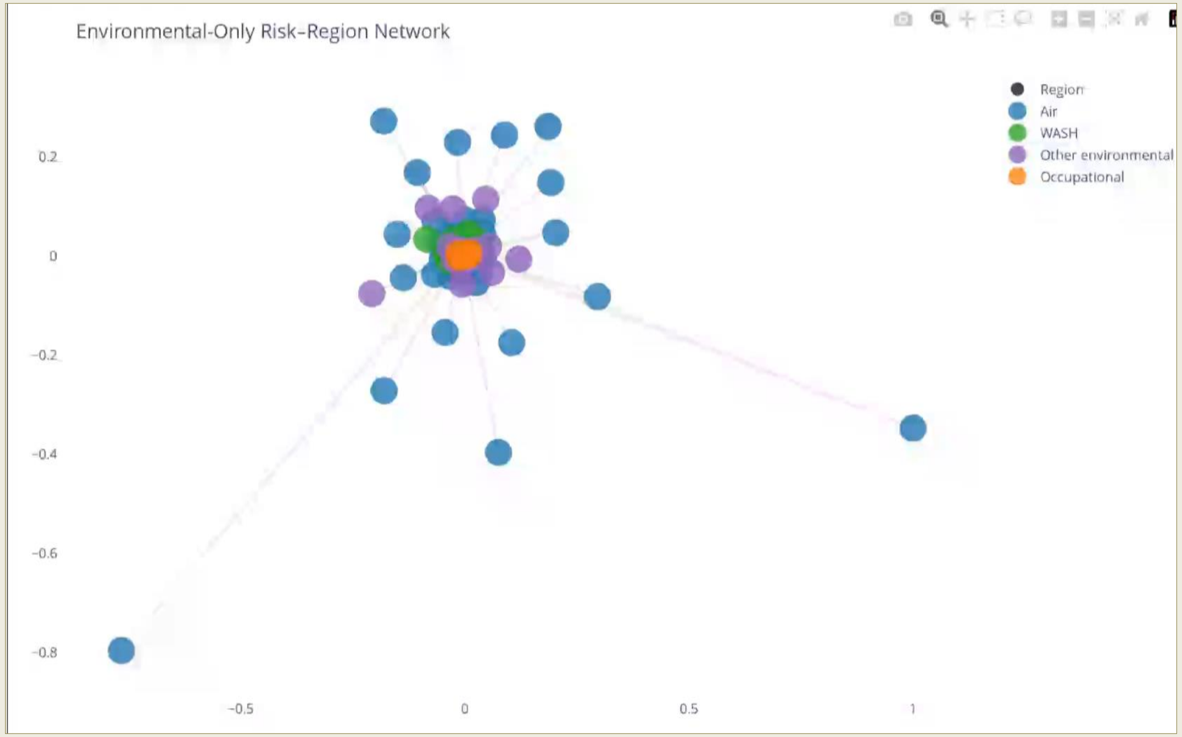
Network Visualization

Shows how environmental risks connect across regions, exposing patterns that help users understand which groups have higher or lower associated risks.

Interactivity

Tooltips:

- Shows groups and distinct risks.



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Key findings

Helps reinforce that air and water pollution are more common than other categories, highlighting the importance of prioritizing these risks.

Conclusions

Global Choropleth Map

Stacked Area Chart

Scatter Plot

Heatmap

Network Visualization

- Environmental risks account for only a fraction of the total global disease burden.
- The trend shows that the global health burden is in decline, including the environmental burden.
- Air and water pollution are the most common environmental risk categories.
- Although some countries are exposed to multiple environmental risks, this does not necessarily mean they have the highest environmental burdens.
- Despite overall declines, some regions still face high environmental burdens, emphasizing the need for continued targeted interventions.
- These patterns reveal persistent global health disparities.

Environmental risks contribute to health problems differently across countries and regions, yet their overall share of the global disease burden remains relatively small. The visualizations and cleaned dataset were essential in addressing the research question by clarifying where these risks are most influential and how they differ across the world.

Gaining a broader understanding of socioeconomic and geographic factors helps place these patterns in context and strengthens efforts to reduce the overall health burden, which is clearly advantageous for improving population health.

THANK YOU

References

Institute for Health Metrics and Evaluation. (2025). *Global Burden of Disease Study 2023 (GBD 2023): Years lived with disability, disability-adjusted life years, healthy life expectancy, and risk-attributable burden 1990–2023*. Institute for Health Metrics and Evaluation. https://ghdx.healthdata.org/record/ihme-data/gbd-2023-yld-daly-hale-risk-1990-2023

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